Zilong Zheng

TU Delft MSc Robotics Student | Seeking an Internship / Graduation Project
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EDUCATION

Delft University of Technology — **MSc in Robotics** — Sep 2025 – Jun 2027

- Research interests: world models, multimodal perception and robot learning

Beihang University — **BSc in Automation** — Sep 2019 – Jun 2024

- GPA: 89.61/100; Rank: 36 / 242

CORE EXPERTISE

- **World Models:** Action-conditioned video representation learning, latent-space dynamics modeling and Hamiltonian constraints
- **Autonomous Robotic Systems:** ROS 2-based perception, SLAM, planning, navigation and Sim-to-Real deployment
- **Motion Planning & Control:** Bi-Informed RRT*, MPPI/MPC and trajectory optimization for 10-DoF robotic systems

EXPERIENCE

Guangxiang (Beijing) Technology Co., Ltd. — **World Model Research Intern** — Jul 2026 – Present

- Compared LDA-1B, Cosmos 3 and V-JEPA 2 in latent video representations, action conditioning; reproduced and evaluated V-JEPA 2 on IntPhys 2 and Diving for latent dynamics, temporal prediction and physical consistency.
- Extended this study by reproducing PH-Dreamer and exploring Hamiltonian structure for latent-state modeling and prediction.

Shanghai Artificial Intelligence Laboratory — **LLM Research Intern** — Dec 2023 – Aug 2024

- Designed and developed MathBench, a hierarchical benchmark for evaluating the mathematical reasoning of large language models; built an automated evaluation pipeline with OpenCompass, evaluated 30+ models and released it through OpenCompass.
- Diagnosed a Russian-language tokenization issue in InternLM2 through cross-lingual token compression analysis and case-level inspection; supported the pre-training team in validating the fix and restoring the compression ratio to the expected range.

Institute of Automation, Chinese Academy of Sciences — **Computer Vision Research Intern** — Jul 2022 – Oct 2022

- Reimplemented core components of ViT and Swin Transformer and ported the UperNet framework from PyTorch to MindSpore.

PROJECTS

ROS 2 Mobile Robot Mapping, Navigation and Sim-to-Real Deployment — Mar 2026 – Jun 2026

- Built an end-to-end autonomy stack covering multi-sensor perception, SLAM, path planning and navigation; integrated TF2, LiDAR, odometry and costmaps, and transferred the system from Gazebo simulation to a MIRTE Master mobile robot.
- Integrated SLAM Toolbox, Nav2 and m-explore-ros2 for autonomous exploration, map reuse and real-time localization; demonstrated closed-loop operation in a 32 m^2 greenhouse and resolved coordinate-frame misalignment, execution latency and narrow-passage navigation failures.

Camera-Radar BEV Fusion for Multimodal 3D Object Detection — Feb 2026 – Apr 2026

- Built a CenterPoint-based multimodal 3D detection framework that jointly models image features and aggregated multi-sweep radar features in bird's-eye-view space.
- Developed a radar-guided BEV enhancement module combining sparse depth cues, multi-sweep radar aggregation and Doppler/RCS features to improve cross-modal spatiotemporal alignment, achieving a 34.8% relative improvement in BEV mAP over the baseline on the View-of-Delft dataset.

Hierarchical Motion Planning for a 10-DoF Robot with RRT* and MPPI — Nov 2025 – Jan 2026

- Built a global-local planning and control framework in MuJoCo and JAX, combining Bi-Informed RRT* for global path search with GPU-accelerated MPPI/MPC for receding-horizon trajectory optimization.
- Designed an MPPI objective incorporating goal tracking, control effort, dynamic constraints and collision penalties, enabling smooth, dynamically feasible and collision-free motion; completed closed-loop simulation, validation and parameter tuning.

PUBLICATION

H. Liu, **Z. Zheng**, et al. "**MathBench: Evaluating the Theory and Application Proficiency of LLMs with a Hierarchical Mathematics Benchmark.**" *Findings of the Association for Computational Linguistics: ACL 2024.*

TECHNICAL SKILLS

Programming & Machine Learning: Python (PyTorch, JAX, NumPy, scikit-learn, MindSpore) | C/C++ | MATLAB
Robotics & Simulation: Linux | Git | ROS2 | Gazebo | OpenCV | MuJoCo | PCL